ShopDesk

Complete Software Requirements Specification

Single Computer / Flutter Windows Desktop / Offline SQLite Database

Consolidated requirements for a multi-product retail shop (mattresses, door mats, chatai, floor paper, table sheets, blankets and other goods).

Platform	Windows 10 / 11 desktop
Database	SQLite — single local computer
Billing	Manual selling prices; automatic stock deduction
Inventory rule	Zero/negative stock must not block sale
Business modules	Products, purchases, sales, parties, expenses, payroll, accounts, ledgers and reports
Delivery target	One connected application and Windows release bundle

Operating flow

Category & custom fields → Product / variant → Stock & purchases → Manual sale → Automatic stock and account posting → Filtered reports → A4 / PDF / Excel

01 | Product Vision & Environment

- One complete Windows desktop application for a single retail shop that sells mattresses, door mats, chatai, floor paper, table sheets, blankets and any future product types.
- Flutter Windows frontend with a modern, responsive desktop interface and one local SQLite database; internet access is not required for daily operation.
- All modules must be accessible inside the same application and ship in one Windows release bundle / installer—not as separate apps.
- Primary objective: simple stock control and manual billing, with linked purchases, parties, ledgers, expenses, employees, accounting and rich reporting.
- Default currency PKR; monetary values stored with exact minor-unit precision. Calendar dates, invoice numbers, transaction timestamps and audit history must be retained.

02 | Dynamic Categories, Products & Variations

- Unlimited categories and subcategories; rename, archive and search categories while preserving historical documents.
- Unlimited products with product names, category, optional SKU/barcode, active/archived state and optional notes.
- Category-specific custom attributes that shop staff can create without changing software code: text, numeric, choice/dropdown, dimension and other valid types.
- Examples: Mattress—size, brand, thickness, density, foam type; Door Mat—color, size, material, design; Chatai—length, width, quality; Floor Paper—roll width, length, design; Blanket—single/double, material and weight.
- Product variations represent individually tracked sellable SKUs; each SKU has its own unit and independent stock balance.
- Attribute choices and validation can be edited; product/variation attributes can be updated safely without rewriting posted invoices.

- Units such as piece, roll, meter, foot, yard, packet and set; unit conversion mapping is required when purchases and sales use different units.
- Fast search by name, SKU, barcode, category, custom field and variation; archived stock remains reportable in history.

03 | Stock & Inventory Ledger

- Opening quantity per variation; stock adjustments with reasons; damage, shrinkage, disposal, returns, transfers (if future locations added) and historical movement journal.
- One source of truth: live stock = opening quantities + purchases + sale returns + positive adjustments - sales - purchase returns - negative adjustments.
- Negative stock is explicitly allowed for sales: saving a bill must NOT fail merely because current stock is zero or less than ordered quantity; display a warning instead.
- Example: opening 0, sell 5 => -5; later purchase 12 => 7. Do not create replacement SKU just because its stock is negative.
- Quantities may have up to three decimal places where the unit supports partial quantities; invalid/NaN quantities must be rejected.
- Atomic posting: sale/purchase, party ledger, journal and stock movements all commit together or all roll back together.
- Stock views: per variation, category, low/zero/negative, stock as of selected date, movement source, period and adjustments.
- Inventory costing policy must be selected and implemented (weighted-average or FIFO) before showing authoritative stock value or gross/net profit, especially when negative stock is allowed.

04 | Sales & Manual Billing

- Manual invoice creation with transaction date, optional customer for walk-in cash sale, multiple product variations, quantity, per-line manually entered price and optional notes.
- Selling price must never be forced from a saved product price; different customers and every bill may have different rates.
- Bill calculates line totals, final total, amount paid now and remaining balance; cash, bank, credit and partial payments are required.
- Credit/partial sale must be linked to a customer; customer receivable updates when bill is posted.
- Save bill even when stock turns negative; post exact negative quantity against the selected SKU.
- Invoice lists, search, date filter, detail view, A4 invoice print, PDF and optional bill export.
- Invoice correction/edit must safely reverse original entries and post corrected invoice atomically, preserving original invoice number in audit history; no destructive silent rewrite.
- Cancellation/void and sales returns must reverse the applicable stock, accounts and party ledgers; returns support item-level partial quantities and refund/credit.
- Optional future enhancements: line discount, invoice discount, tax and configurable receipt formats. These must not be assumed as complete until tested.

05 | Purchases & Suppliers

- Supplier register: name, contact, phone, address, opening balance, active/archived status, purchase history and outstanding amount.
- Purchase invoice supports supplier, date, multiple SKUs, quantity, manually entered unit cost, total, paid now, due and reference.
- Confirming a purchase increments stock for exactly the selected SKU; purchases repay a negative stock balance automatically.
- Supplier credit/partial payment increases payable balance; later payments reduce payable balance.
- Supplier return supports original purchase line, partial return quantity, stock reduction, refund and ledger reversal.

- View/print/export supplier ledger with dated debits, credits and running balance; payment history, adjustments and opening balances.
- Purchase corrections/void must recalculate quantities and supplier payable without duplicates; keep audit trace.

06 | Customer Register & Receivables

- Customer register with name, phone, address, optional notes, opening balance, paid/credit history and statement.
- Customer detail offers View Ledger, Add Payment, Payment History, Edit, Archive and A4/PDF/Excel export.
- Customer receivable = total customer debits minus credits; advance payments/credit balances remain visible and distinct from amounts due.
- Report customer balances as of selected date; show running balance, transaction references, invoice links, receipts and refunds.
- Receipt or payment edit/cancel must reverse its posting with an auditable entry; not silently delete linked history.

07 | Expenses & Categories

- Unlimited expense categories such as rent, utilities, transport, maintenance, wages and miscellaneous; rename and manage categories.
- Expense entry: date, category, amount, cash/bank account, description and optional reference/receipt attachment.
- Expense report must filter by date, category, payment account, amount range, text search and transaction status.
- Every approved expense posts to expense/cash or bank accounts. Corrected or cancelled expenses reverse related journal entries with audit history.

08 | Employees, Salaries, Advances & Ledger

- Employee details: full name, designation/role, start date if provided, monthly salary rate, active/archived status and notes.
- Advance payment made to an employee is saved immediately against both employee advance ledger and selected cash/bank account.
- Payroll by employee and YYYY-MM period: gross salary, outstanding advance, recovery amount, other approved deductions, pay now and unpaid salary.
- Advance recovery cannot exceed either available advance or salary; unadjusted advance carries forward.
- Part-pay or defer salaries; later payment of outstanding salary reduces Salary Payable and cash/bank balance.
- One employee ledger showing advances, recoveries, gross salary dues, paid amounts, adjustments and running advance/salary balances.
- Protect against duplicate payroll for the same employee/month; allow controlled correction/reversal with an audit trail.
- Employee register, salary report, advance report and employee ledger all export/print in A4/PDF/Excel.

09 | Financial Accounts & Double Entry

- Chart of accounts: Cash, Bank, Accounts Receivable, Inventory, Employee Advances, Accounts Payable, Salary Payable, Equity, Sales Revenue, Cost of Sales, Operating Expenses and Salary Expense.
- Cash/bank payments, transfers, receipts and balanced manual journal entries with references and dates.
- Sales, purchases, returns, receipts, payments, expenses, employee advances and payroll produce balanced accounting postings in the same transaction as source records.

- Opening balances for customers, suppliers, cash/bank, inventory and chart of accounts with proper balancing equity entries.
- Trial balance, account ledger, cash/bank book, revenue/expense summary and reconciliation controls.
- Full profit-and-loss, inventory valuation and balance sheet require complete and verified inventory COGS and closing rules; they must never be labelled final based on revenue minus operating expenses alone.
- Posted transactions must be voided/reversed rather than hard-deleted; account numbers and historical journals must remain immutable.

10 | Reports — Mandatory Catalogue

- Sales report: invoice, item, category, variation, customer, payment status, date, amount and totals.
- Purchase report: supplier, item, category, variation, cost, paid/due, date and totals.
- Customer list and Customer Ledger (running balance), Customer Receivables / outstanding aging, customer payments.
- Supplier list and Supplier Ledger (running balance), Supplier Payables / outstanding aging, supplier payments.
- Expense summary and category-wise expense reports, employee register, employee ledger, payroll, salary outstanding and advances.
- Stock by product/category/variation, negative/zero stock, stock movements, stock as of selected date, opening and stock adjustment history.
- Product and variation register; sales returns, purchase returns, account ledger, cash/bank report, trial balance, daily/monthly sales, audit trail.
- Profit & Loss, Balance Sheet, inventory valuation and COGS become deliverables only after cost accounting acceptance is completed.

11 | Reports — Filters, Export & Printing

- All reports offer only meaningful filters from: custom start/end dates, today/week/month/year, product, SKU variation, category, customer, supplier, employee, expense category, account, cash/credit status, negative/zero/positive stock, invoice number, search text and min/max amount.
- Group/sort when meaningful by day, month, product, category, customer, supplier, account or payment status; no fabricated groupings for incompatible reports.
- Filter controls are dynamic for each report; changing filters must regenerate preview and totals, not merely hide rows in a table.
- Export the filtered results to native Excel .xlsx and PDF; use A4 page formatting for PDF and Windows print.
- Portrait for narrow reports, landscape for wide reports; repeated column headers, page numbering, consistent company heading, date range, readable currency and safe row pagination.
- Preview can paginate large reports but export must include ALL filtered rows. Numeric columns remain numeric in Excel if feasible.
- Customer, supplier and employee detail screens provide direct ledger export and print shortcuts.

12 | UI/UX & Accessibility

- Professional Flutter desktop interface: single consistent dark navigation rail/sidebar, light content workspace, search, dashboards, forms and tables.
- Consistent input validation, spacing, typography, confirmation dialogs and error messages across every module.
- Desktop-friendly keyboard and mouse interaction: tab order, keyboard shortcuts where practical, sortable table headers, focus indicators, scrollable wide tables.
- Never create an apparently working button without implementation; error messages must explain the action and preserve entered form state where possible.

- Dashboard tiles for today sales, total SKUs, zero/negative stock, customer receivables and supplier payables; do not label revenue as profit.
- Business-critical actions (void/refund/return/correction/restore) need confirmation and clear consequences.

13 | CRUD, Audit & Data Integrity

- Create, read, update and archive for product groups, products, variations, customers, suppliers, employees, expense categories and user-editable settings.
- All posted transactions support auditable correction/void/return with compensating entries; hard delete of financial records is prohibited.
- Stock and general-ledger journals use foreign keys, constraints, prepared query arguments and atomic SQLite transactions.
- Reject invalid amounts, non-finite numbers, incorrect parties, invalid date formats and excess returns; do not reject valid negative stock.
- Maintain audit timestamps and old/new record identifiers for corrections; preserve historical reports and transaction references.
- SQLite foreign key checks, integrity_check, journal debit-credit equality and returns quantity controls available in a diagnostics screen.

14 | Backup, Restore, Security & Packaging

- Single-PC offline storage in Windows application-support directory, not inside a temporary unzip folder.
- One-click consistent SQLite snapshot backup, date-stamped filenames and clear data location.
- Validated restore process while app is closed; preserve previous database and test restored checksum/integrity prior to replacement.
- Automatic backup schedule and off-device copying recommended; encryption and sign-in/access controls specified for hardening when live finance records will be stored.
- Generate one Windows release build containing all Flutter modules. Include runtime DLLs/data, installation instructions and desktop shortcut / installer if requested.
- All scripts/CI must run flutter create --platforms=windows ., flutter pub get, flutter analyze, flutter test and flutter build windows --release; release only if those steps pass.
- Provide user-facing installation, operation, backup and troubleshooting manual; production rollout requires trial with copy of real shop stock and balances.

15 | Core Workflow Acceptance Tests

- AT-01: Create new category with user-defined size/thickness/brand fields, then create two SKUs with different attributes and independent opening quantities.
- AT-02: With zero stock, sell 5 SKU units at user-entered unit price: bill saves and stock becomes -5, with linked customer ledger if credit.
- AT-03: Purchase 12 of same SKU from supplier: stock becomes +7; supplier payable and journal reconcile.
- AT-04: Correct posted invoice quantity/price or party: old invoice voided, new invoice posted atomically; failed correction leaves original untouched.
- AT-05: Partially return sold goods; exact line quantity restored; customer refund/credit and journal updated, no excessive returns possible.
- AT-06: Partially return purchased goods; correct stock reduction and supplier refund/credit.
- AT-07: Customer buys on credit, makes multiple receipts and has a running ledger with correct opening, debit, credit and current balance.
- AT-08: Supplier purchases on credit then is paid partially; supplier payable drops by payment amount without changing stock.
- AT-09: Enter and reverse expense; expense-category report and cash/bank journal match.

- AT-10: Employee takes advance, payroll recovers some, unpaid salary is settled later; employee advance and salary due are correct.
- AT-11: Apply date/category/product/variant/customer/supplier/employee/payment-status and negative-stock report filters; result counts and totals match raw records.
- AT-12: Export long/wide reports to Excel/PDF and print A4 with repeated headers; all filtered rows present.
- AT-13: Verify SQLite integrity, all journal postings balanced, and original posted transaction history preserved after correction/void.
- AT-14: Back up database, restore in a safe test location, reopen and compare customer balances, supplier balances and stock against source.
- AT-15: Build and launch the Windows release on the target computer; test full purchase -> sale -> payment -> return -> report workflow.

16 | Delivery Definition of Done

- The delivered release is ONE Windows desktop application with all required modules wired to one durable SQLite database.
- Every module has tested create/edit/archive (masters) or post/correct/void/return (financial) flows and helpful error states.
- Negative stock behaviour remains enabled and does not block a valid invoice.
- Full report/filter matrix supports tested Excel export, PDF and A4 printer layout.
- Financial valuation, closing and profit-related pages are either verified or explicitly disabled and labelled incomplete.
- Automated Flutter tests, Windows analyze, Windows release build, installer smoke test and sample-data acceptance checklist are completed without errors.
- Shop staff receive an operational walkthrough, backup/restore procedure and a method to report any bugs before using live accounts.

Report Filter Matrix

Report family	Applicable filtering / grouping
Sales / billing	Date, customer, product, SKU, category, payment status, range, invoice
Purchases	Date, supplier, product, SKU, category, payment status, range
Stock / negative / as-of	As-of date, product, SKU, category, negative/zero/positive
Stock movements	Period, product, SKU, category, source
Customers / receivables	Customer, as-of date, name search, aging
Customer ledger	Customer, date range, running balance, transaction reference
Suppliers / payables	Supplier, as-of date, aging, name search
Supplier ledger	Supplier, date range, running balance, transaction reference
Expenses	Date range, category, cash/bank account, amount range, search
Employees / payroll	Employee, period, dues, advance, role
Employee ledger	Employee, period, entry type
Accounts / cash bank / trial balance	Account, date range, reference, account type
Products & variants	Category, product, SKU, custom values
Returns, receipts & audit	Date, reference, party, status, employee where relevant

Implementation Status — Read Before Live Use

The source project is an integrated development version and has NOT been compiled or verified on Windows in this environment. The spec above is the full required scope—not a certificate of complete implementation. Production release is conditional on every mandatory acceptance test, successful Windows build and verified backup restoration.

Outstanding critical engineering work includes full inventory costing/COGS and profit verification, robust period closing, production security and recovery controls, payroll editing/reversals, and physical A4/Excel/printer acceptance on Windows.

Release Readiness Checklist

Area	Current status	Required before live use
Flutter analyzer + tests	Not executed here	Run on Windows and fix all compile/test failures
Windows release / installer	Not built here	Build app and verify all runtime DLLs
Stock costing / COGS	Pending	Implement cost policy incl. negative-stock purchases
Unit conversion	Pending	Support roll-to-meter and unit conversion bookkeeping
Financial period closing	Pending	Restrict changes to posted/closed periods
Permissions / sign-in	Pending	Add appropriate operator access controls
Backup and restore	Partially available	Prove restore and schedule off-device backups
Payroll corrections	Partially available	Add auditable payroll void / editing
A4 report & printer QA	Not verified here	Validate PDF, Excel and print on shop printer
Business acceptance	Not executed here	Pass AT-01 through AT-15 with realistic data